The Urea Cycle Research Hub: Automated Literature Curation and Multi-Parameter Variant Analysis Across Eight Disease Genes

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Abstract

Urea cycle disorders (UCDs) are rare inherited metabolic diseases caused by pathogenic variants in genes encoding urea cycle enzymes and transporters. Interpreting the clinical significance of the thousands of known variants remains a major challenge. We developed the Urea Cycle Research Hub, a web application that automates PubMed literature retrieval across the eight urea cycle genes and includes analysis of AlphaMissense pathogenicity scores, DynaMut2 thermodynamic stability predictions (ΔΔG), ClinVar classifications, REVEL ensemble scores, and structural context from experimental crystal structures. Across the eight genes, 2,341 unique missense variants were catalogued. AlphaMissense score distributions varied substantially by gene, with OTC exhibiting the highest median and NAGS the lowest. In OTC, AlphaMissense scores were significantly associated with ClinVar classifications ($p = 1.58 \times 10^{-23}$) and negatively correlated with experimental functional scores from a 1,570-variant yeast growth assay by Lo et al. ($r = -0.446$, $n = 376$). DynaMut2 ΔΔG values showed a weaker but significant association with the same data ($r = 0.286$, $n = 373$). Concordance analysis revealed that 59.5% of OTC variants exhibited high AlphaMissense scores despite minimal thermodynamic destabilization, suggesting that active site disruption is a predominant pathogenic mechanism. This is

reinforced by a strong association between active site proximity and predicted pathogenicity (χ^2 $p = 1.24 \times 10^{-16}$). Validation against 18 kinetically characterized variants confirmed strong agreement for severely impaired substitutions while identifying limitations for moderately reduced variants. This platform integrates computational and structural features in an accessible community resource to prioritize urea cycle variants for experimental validation and clinical interpretation.

Introduction and Background

The urea cycle is the primary pathway for ammonia detoxification in humans, converting toxic nitrogen produced from amino acid breakdown into urea, and serves as the principal mechanism for the clearance of waste nitrogen from protein and other nitrogenous compounds. Additionally, this pathway also produces arginine, ornithine, and citrulline as vital intermediates—key amino acids used in metabolic signaling (Simpson *et al*., 2025). Deficiencies in Urea Cycle enzymes lead to several urea cycle disorders (UCDs), and these can be grouped as metabolic errors characterized by hyperammonemia and neurological impairment. The overall incidence of UCDs in live births is measured to be around 1 in 35,000, making them a rare but significant condition (Häberle *et al*., 2019). Among these, ornithine transcarbamoylase deficiency (OTCD), an X-linked UCD, is the most common, often described as “the most common ureagenesis disorder in the world” (Micheloni *et al*., 2025), where patients suffer variable levels of hyperammonemia depending on the severity of the deficiency.

UCDs arise from pathogenic mutations in eight genes of the Urea Cycle Pathway: six enzymes (OTC, CPS1, ASS, ASL, ARG1, NAGS) and two mitochondrial transporters (ORNT1 and CITRIN) (Simpson *et al*., 2025). Inherited deficiencies in any of these genes can disrupt the ammonia detoxification process, leading to metabolic crises and, if left untreated, neurological damage or death.

Hundreds of distinct variants between these eight genes have been documented across the literature. For example, by 2006, 341 unique OTC variants were known, and by 2015, this number increased to 417 disease-causing variants, most of which were single-nucleotide substitutions (Caldovic *et al*., 2015). Similar levels of diversity can be observed in the other metabolic genes in the Urea Cycle, and in all cases, both private, family-specific mutations and

recurrent mutations occur (Yamaguchi *et al.,* 2006). Recurrent mutations are better characterized due to the large number of cases, but private mutations typically arise as a *de novo* mutation in a single lineage, causing them to be harder to characterize as pathogenic or benign. This complicates the interpretation of variants, as many variants remain unclassified due to the lack of structural data.

Thus, experimental characterization is limited. Micheloni *et al*. investigated 17 OTC variants through kinetic assays and bioinformatic analysis, and revealed that catalytic site coupling was correlated with enzyme impairment. More recently, Lo *et al.* expanded this work by developing a high-throughput assay that measured the functional impact of 1,570 individual amino acid substitutions in human OTC that covered 84% of all possible missense mutations. Their results showed that large-scale functional assays can effectively distinguish benign from pathogenic variants and even differentiate neonatal-onset from late-onset disease presentations (Lo *et al.*, 2023). However, most mutations don't undergo such a detailed analysis, creating a gap between the discovery of genetic mutations and their functional interpretation. While computational tools such as PolyPhen, SIFT, and AlphaMissense provide probabilistic assessments, they are still considered "supporting evidence" rather than a guaranteed predictor of pathogenicity or benignancy. As *Nature* writer Ewen Callaway states, "computational predictions currently have a minimal role in diagnosing genetic diseases... [and] tools such as AlphaMissense will need to undergo thorough testing before they are used in the clinic" (Callaway, 2023). Nonetheless, their structural predictions can still provide valuable insights into the potential functional impact of these mutations.

Due to the vast amount of data available on UCDs and specific genes, automated literature retrieval is essential to maintain an up-to-date view of gene-specific research. To

accomplish this, we developed a web application that continuously queries the NCBI databases (PubMed, PMC) using the Entrez Programming Utilities (E-utilities) to retrieve literature related to a set of key terms, in this case, the specific urea cycle genes.

The objective is to create an easily accessible knowledge base of available literature, providing clinicians and researchers with a common resource. In addition, this enables the creation of agents that can inform clinicians who do not regularly encounter patients with urea cycle disorders, allowing them to prioritize specific variants for further study. Variant data was further built upon by incorporating submissions from the ClinVar database (Landrum et al., 2018), a community-curated repository of clinically observed variants with established, agreed upon pathogenicity classifications. To connect variants directly to the literature where they are described, the LitVar 2.0 system (Allot et al., 2023) was deployed, allowing retrieval of PubMed identifiers linked to specific variant queries. Using LmRaC, Craig and Drăghici have further modified this type of retrieval system, which "allows users to dynamically build domain-specific knowledge bases from PubMed sources… answers are drawn solely from this RAG [retrieval-augmented generation] with citations to the paragraph level" (Craig and Drăghici, 2024). In other words, an LLM that only answers questions using relevant PubMed paragraphs as a domain. Our approach focuses on a specific biological domain, the Urea Cycle, and permits dynamic data retrieval and variant extraction rather than functioning simply as an interactive query tool.

Furthermore, PubTator 3.0, developed by Wei *et al*., shows the usage of AI-powered text mining in literature. The tool "provides advanced search capabilities and enable[s] large-scale analyses, streamlining many complex information needs… [it] allow[s] researchers to navigate the ever-expanding wealth of biomedical literature" (Wei *et al*., 2024). By integrating this approach, our system can find patterns and automatically extract, annotate, and organize

publications by gene and variant, creating a dynamic research dashboard that can be utilized by UCD investigators.

In combination with literature mining, computational modeling was applied to the identified and cataloged mutations to obtain a functional interpretation. AlphaFold-based models are used in this project, not for diagnostic purposes but to prioritize variants for further biochemical or experimental validation. Building on AlphaFold, Cheng *et al*. developed AlphaMissense, a software specifically designed to predict missense variant pathogenicity with highly accurate results (Cheng *et al*., 2023). Unlike the AlphaMissense model, which applies deep learning to predict pathogenicity across the entire human proteome, this project focuses on targeted Urea Cycle genes and integrates variant data with AlphaFold structural modeling. To complement the sequence-based prediction, thermodynamic stability upon mutation (ΔΔG, kcal/mol) was estimated using DynaMut2, a web server that combines Normal Mode Analysis with graph-based signatures to predict the effect of missense variants on protein stability and dynamics (Rodrigues et al., 2021). The aim is not to train a new predictive model, but to contextualize existing mutations within biochemical and structural frameworks.

The Urea Cycle Research Hub was developed to allow a community of researchers and clinicians to access a unified knowledge base of urea cycle literature, contextualize computationally scored variants, and interact with three-dimensional structural data within a single web interface. Across all eight urea cycle genes, 2,341 unique missense variants were catalogued through both literature extraction and ClinVar import, with AlphaMissense, REVEL, DynaMut2 scores, and ClinVar classifications integrated for each. The automated literature retrieval system in conjunction with a multi-parametric structural and computational analysis shows the integrated approach required to advance variant interpretation in rare metabolic

diseases, providing a viable and accessible platform for the UCD research community.

Materials & Methodology

As the first part of this project, the Urea Cycle Research Hub is a Flask-based web application (Flask 3.1.2, Python 3.12.0) designed to automate the retrieval, curation, and management of urea cycle research publications. The frontend was built using HTML5, CSS3, and JavaScript, and a SQLite database was also added for more efficient publication storage. This database holds researcher opinion data through an upvote/downvote system and user-generated comments and also stores keywords that users can use for custom searches. The final application will be deployed with support for Docker 24.0.6 containerization and cloud platform hosting as well.

The primary feature of the site is the display of PubMed query results, complete with bibliographic metadata (title, abstract, journal name, publication year, volume, page numbers, PubMed ID, PubMed Central ID, and full publication dates). Publication data was taken from the National Center for Biotechnology Information (NCBI) PubMed database using Biopython 1.86, specifically the Entrez module. The site uses search queries modified for each of the eight urea cycle genes and transporters: OTC, CPS1, NAGS, ASS, ARG1, ASL, ORNT1, and CITRIN. Queries use Boolean logic in the format ("urea cycle" OR "ureagenesis") AND (gene_term1 OR gene_term2) AND (user_keyword1 OR user_keyword2), etc, with multi-word keyword phrases enclosed in quotation marks to accurately match PubMed results. Additionally, users can also create their own custom keyword combinations and upload their searches into the community page for other researchers to experiment with. The paper lists at each gene page (or custom search), can be sorted chronologically (publication year) or by popularity (like/dislike count). Over time, community filtering becomes more efficient as more papers are reviewed by users and their inputs logged.

Anonymous user tracking was implemented using a cookie-based identification system where each user receives a universally unique identifier (UUID) stored in a browser cookie with a 10-year expiration period upon first visit. This approach tracks the previous interactions of users without requiring authentication, leading to greater convenience for community participation. As stated before, users can express approval or disapproval for individual publications through like and dislike buttons, add comments to any publication, and define custom keywords that are automatically saved in the database and restored upon return. A cookie-based tracking system also records the timestamp of each user's last visit to specific gene pages and notifies users of newly published papers since their last visit. A dedicated community page lets urea cycle researchers share effective keyword search combinations with descriptive topic labels and comma-separated keywords, which can be liked and commented on by other users in the same way.

An automated email notification system was also implemented using Flask-Mail 0.9.1 to deliver weekly publication updates. Users can subscribe via email and customize their subscriptions at the gene level (OTC, ARG1, etc) or by creating custom keyword-based subscriptions. Email generation is done through a weekly cron job that executes the flask send-weekly-emails CLI command. The system searches through PubMed for new publications that match each user's selected subscriptions every Monday morning, and filters results to include only papers published since the last email sent.

For the lists at each gene page (or custom search), data export in RIS format is available (compatible with EndNote, Mendeley, and Zotero). The RIS export contains the same metadata shown in the site, but also includes the scientists’ opinions with likes, dislikes, and comments. This lets researchers import this community-curated data for their own purposes, such as

reference libraries. Raw PubMed ID list exports in plain text format are also available, with one PMID per line.

Variant Extraction and Database Integration

Variants were extracted from the PubMed abstract cache with two text-mining approaches. First, a regular expression pattern matching standard HGVS protein notation (e.g., [A-Z][0-9]{1,4}[A-Z], with variants such as R40H) was applied to all cached abstracts for each of the eight variant genes, along with a curated exclusion list filtering known false-positive patterns such as diagnostic abbreviations. Second, the PubTator 3.0 API (Wei et al., 2024) was queried using the PMIDs of cached papers to retrieve variants, which were merged with regex results and deduplicated. Each extracted variant was stored with its source gene, amino acid substitution in short form, source PMID, and extraction method (regex or pubtator).

Variants were also imported from ClinVar (Landrum et al., 2018) through the download of gene-specific tab-delimited missense variant exports. These were imported into the same variants table with the extraction method label “clinvar” to separate them from literature-derived variants. ClinVar classification terms (Pathogenic, Likely Pathogenic, Variant of Uncertain Significance, Likely Benign, Benign) were retained as a separate field and displayed on variant detail pages as the sole clinical designation.

AlphaMissense Score Integration

AlphaMissense scores for all possible human missense variants were downloaded from the resource included in Cheng et al. (2023) as a tab-separated values file. Rows corresponding to the eight urea cycle genes were filtered by UniProt accession and imported into the database,

indexed on gene, amino acid position, reference amino acid, and alternate amino acid. Scores range from 0 to 1; higher values indicate greater predicted functional impact, or pathogenicity. Following the recommendation of the original authors, scores are displayed as numeric values without pathogenicity classification, acknowledging the limits of computational predictions.

Structural Data Acquisition

Experimental crystal structures were obtained from the RCSB Protein Data Bank (Berman et al., 2000) for each gene where a suitable human structure was available (Table 3). Ligand-bound structures were preferred over apo forms to make active site identification much simpler from the coordinates of bound substrates or inhibitors. For OTC, the 1C9Y structure (1.85 Å) was used, which includes the competitive inhibitor phosphonacetyl-L-ornithine bound in the active site. For CPS1, the ligand-bound 5DOU structure (2.6 Å) was selected over the apo form 5DOT. For ASL, the 1AOS structure (4.2 Å) was used; but given the low resolution of this structure, structural annotations for ASL variants are flagged as lower-confidence in the site. For ORNT1 and CITRIN, no suitable experimental human structures were identified, so predicted structures from the AlphaFold Protein Structure Database (Jumper et al., 2021) were used instead. All structure files were stored locally within the application and served to the Mol* viewer.

Table 3: Crystal structures used

Gene	UniProt ID	PDB ID	Resolution (Å)	Ligand(s) Bound	Structure Source	Notes
OTC	P00480	1C9Y	1.85	PALO (inhibitor)	RCSB PDB	Inhibitor-bound; preferred reference structure
CPS1	P31327	5DOU	2.60	ADP, NLG, MG,	RCSB PDB	Ligand-bound form; preferred over apo 5DOT
NAGS	Q8N159	4K30	N/A	—	RCSB PDB	Fill resolution from RCSB if known
ASS1	P00966	2NZ2	N/A	—	RCSB PDB	Fill resolution from RCSB if known
ASL	P04424	1AOS	4.20	None	RCSB PDB	Low resolution; side-chain annotations flagged as lower-confidence
ARG1	P05089	2PHA	N/A	—	RCSB PDB	Fill resolution from RCSB if known
ORNT1	Q9Y619	—	—	—	AlphaFold DB	No suitable human crystal structure identified
CITRIN	Q9UJS0	—	—	—	AlphaFold DB	No suitable human crystal structure identified

Table 3*. Crystal structures used for structural annotation and DynaMut2 ΔΔG calculations. PALO = phosphonacetyl-L-ornithine; NLG = N-acetylglutamate.*

DynaMut2 Thermodynamic Stability Analysis

Free energy changes upon mutation (ΔΔG, kcal/mol) were calculated for extracted variants using DynaMut2 (Rodrigues et al., 2021). Submissions were automated via the DynaMut2 REST API using the batch prediction endpoint (/api/prediction_list), which accepts a PDB file and a text file of mutations in standard notation (one per line, e.g., R40H) and returns a JSON array of per-variant ΔΔG predictions. Each gene's preferred crystal structure was submitted with the complete list of extracted variants for that gene, using chain A in all cases. Positive ΔΔG values indicate predicted thermodynamic destabilization relative to the wild-type structure, and lower values indicate more thermodynamic stability.

Structural Annotation and Active Site Analysis

Active site residues for each gene were gathered from UniProt functional annotations and published biochemical characterization studies. For OTC, active site positions were defined as R92, R141, Q171, D263, S267, and R330 based on Watson et al. (2024) and Micheloni et al. (2025). For remaining genes, residues labeled “Active site” or “Binding site” in UniProt were used. Crystal structure coordinates were parsed using the BioPython PDB module (Cock et al., 2009). For each variant position present in the structure, the minimum Cα–Cα distance to any annotated active site residue was recorded. Secondary structure assignment and relative solvent accessibility (RSA) were obtained using BioPython’s DSSP interface, with residues with RSA below 0.25 classified as buried. All annotations were cached in the database and displayed on variant detail pages. A final visualization was implemented using the Mol* viewer (Sehnal et al., 2021).

Statistical Analysis and Validation

Pearson and Spearman correlation coefficients between AlphaMissense scores, DynaMut2 ΔΔG values, and Lo et al. (2023) functional scores were calculated using scipy.stats (Python 3.12, scipy 1.11). The Lo et al. dataset was obtained from supplementary Table S7 of Lo et al. (2023), containing relative yeast growth scores for 1,570 OTC amino acid substitutions. Variants were matched between datasets on the short amino acid substitution identifier (e.g., R40H). The correlation between AlphaMissense score distributions and ClinVar classification groups was done by the Kruskal-Wallis H-test. The relationship between active site distance bin and AlphaMissense score category was done via a chi-squared test. All figures were generated in Python using matplotlib 3.7 and seaborn 0.12.

Concordance Classification

Variants with both AlphaMissense and DynaMut2 scores were classified into four concordance categories using established thresholds: AlphaMissense > 0.564 (the likely pathogenic boundary defined by Cheng et al., 2023) and ΔΔG > 1.0 kcal/mol (thermodynamically destabilizing threshold). Categories were: both high (AM > 0.564 and ΔΔG > 1.0), both low (AM < 0.34 and ΔΔG < 1.0), AM high / ΔΔG low, and ambiguous (all remaining variants not meeting either pair of thresholds).

Results

Variant extraction for the eight urea cycle genes was done by combining text mining of PubMed-cached literature and structured import from ClinVar. In total, 2,341 unique missense variants were identified and catalogued across OTC, CPS1, ASS, ASL, ARG1, NAGS, ORNT1, and CITRIN (Figure 1).

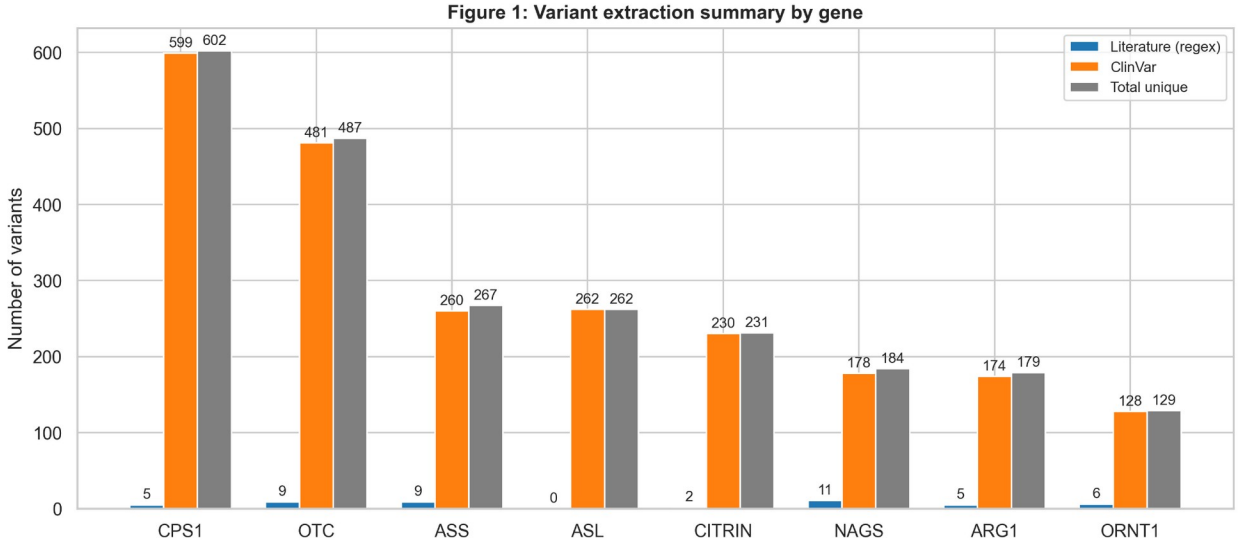


Figure 1. *Variant extraction summary across eight urea cycle genes. Bar chart showing the number of variants identified per gene through literature extraction (regex/PubTator, blue) and ClinVar import (orange), with total unique variant counts (gray). CPS1 yielded the largest variant set (n=602), while ORNT1 had the fewest (n=129). 2,341 unique missense variants.*

CPS1 yielded the largest total variant set with 602 unique loci, followed by OTC with 487 and ASL and ASS with 262 and 267, respectively. ORNT1 had the fewest catalogued variants at 129. Across all genes, ClinVar import was the largest source of variants, accounting for the majority of identified variants for every gene. Literature-based extraction via regex and PubTator contributed a smaller but complementary set: NAGS yielded the highest proportion of literature-only variants (n=11), while ASL yielded none from text mining alone (n=0). The small number of literature-extracted variants is expected, however, given that abstract-level extraction

captures only variants named in PubMed abstracts, and many variants only appear in full text or supplementary materials. This suggests the integration of LitVar as a complementary retrieval pathway to recover additional variants that can't be detected by abstract regex alone. Full per-gene counts, AlphaMissense coverage statistics, and median scores can be seen in Table 1.

Table 1: Per-gene variant summary

Gene	Total Unique Variants	Literature-Extracted	ClinVar-Imported	AM Scored (n)	Median AM Score	% AM > 0.564
CPS1	602	5	599	601	0.418	43.9
OTC	487	9	481	484	0.724	57.0
ASS	267	9	260	264	0.702	55.7
ASL	262	0	262	262	0.474	44.7
CITRIN	231	2	230	229	0.428	42.4
NAGS	184	11	178	181	0.161	19.9
ARG1	179	5	174	171	0.445	42.1
ORNT1	129	6	128	129	0.556	49.6
TOTAL	2341	47	2312	2321	0.473	46.2

Table 1. *Variant catalogue summary across eight urea cycle genes. Literature-extracted variants were identified by regex pattern matching and PubTator 3.0 API annotation of PubMed abstracts. ClinVar-imported variants were obtained from gene-specific missense exports. AM = AlphaMissense.*

AlphaMissense pathogenicity scores were retrieved for all variants with matched amino acid substitution records, with sizes ranging from n=129 (ORNT1) to n=601 (CPS1) across the eight genes (Figure 2). Score distributions varied greatly between genes, demonstrating differences in the possible substitutions that structurally impact each enzyme.

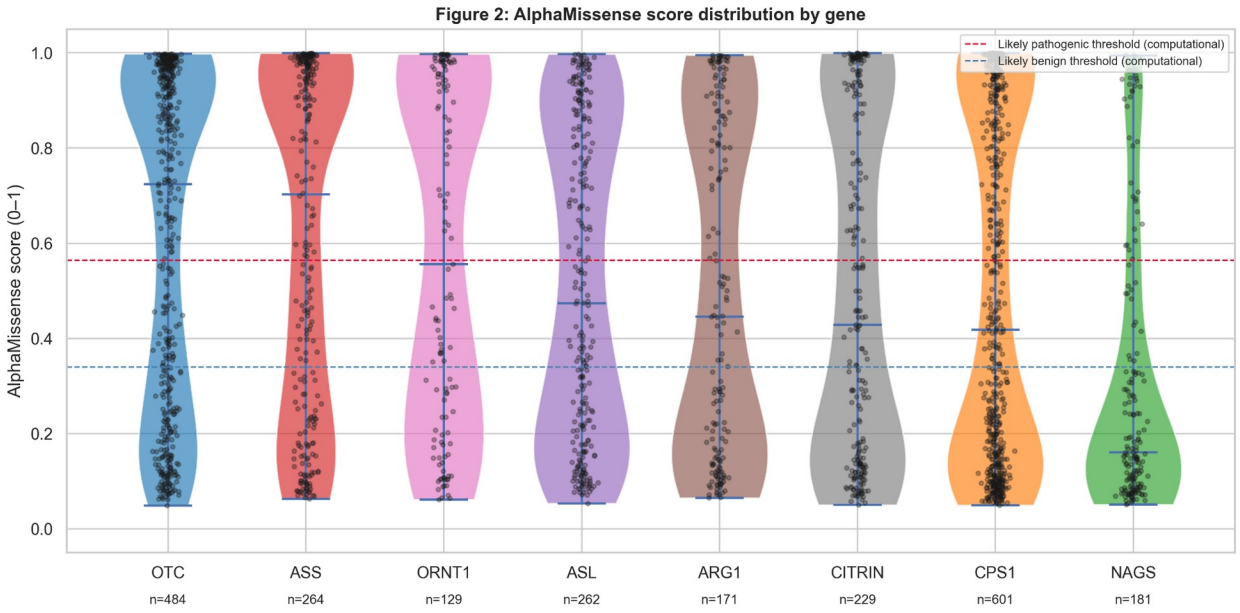


Figure 2. *AlphaMissense score distributions across eight urea cycle genes. Violin plots show the full score distribution for each gene, overlaid with individual variant points (strip plot, α=0.3). Genes are ordered by median AlphaMissense score in descending order. Horizontal dashed lines indicate the computational thresholds for the likely pathogenic (red, AM=0.564) and likely benign (blue, AM=0.34) categories as set by Cheng et al. (2023).*

OTC had the highest median AlphaMissense score of the eight genes (median ≈ 0.73), which is consistent with its known intolerance to missense mutations and is the most commonly mutated urea cycle gene. ASS showed a high median as well (≈ 0.70). In contrast, NAGS displayed a very low median score (≈ 0.17), with the majority of its distribution falling below the 0.34 computational threshold. This suggests that a larger proportion of possible NAGS substitutions are predicted to be functionally neutral. CPS1, the largest enzyme in the pathway, showed a median of approximately 0.42, staying in the middle of the distribution range.

To determine whether AlphaMissense scores correspond to clinical variant classifications, OTC variants with both a ClinVar classification and an AlphaMissense score were analyzed (Figure 3). Pathogenic variants (n=197) had elevated AlphaMissense scores with

a median near 0.90, and likely pathogenic variants (n=66) showed an overlapping distribution with a similarly high median of 0.92. Benign and likely benign variants (n=3 and n=7) showed lower scores with medians near 0.20. Variants of uncertain significance (VUS, n=199) showed the widest distribution, spanning almost the entire 0–1 range with a median of approximately 0.28, reflecting their idiopathic nature.

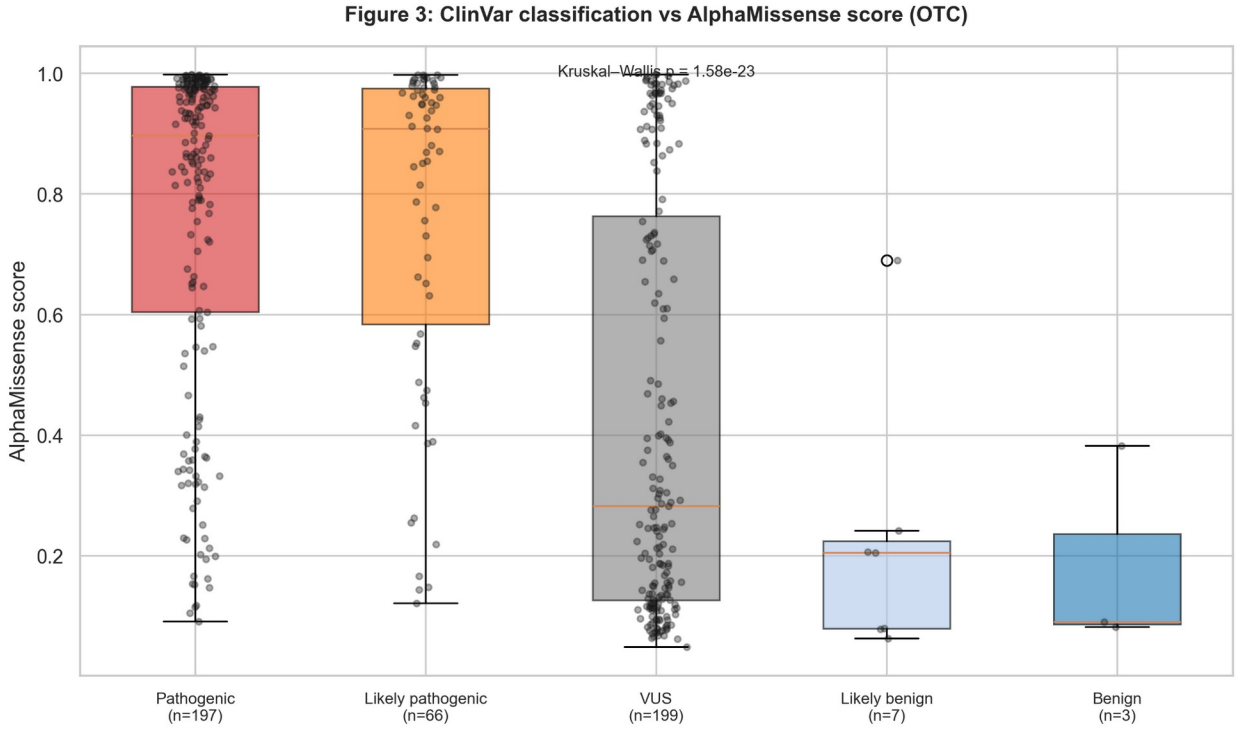


Figure 3. *AlphaMissense score distributions against ClinVar clinical classification in OTC. Boxplots with overlaid strip plots show AlphaMissense scores for OTC variants grouped by their ClinVar designation. Classifications are ordered from most pathogenic (left) to most benign (right).*

A Kruskal-Wallis test confirmed that the difference in AlphaMissense score distributions across the five ClinVar classes was statistically significant ($p = 1.58 \times 10^{-23}$). These results indicate that AlphaMissense scores are associated with established clinical classifications in OTC, while also highlighting the apparent heterogeneity of the VUS category.

To evaluate predicted performance against experimental results, AlphaMissense scores for OTC variants were compared to the relative yeast growth scores from Lo et al. (2023), which measured the functional impact of 1,570 OTC amino acid substitutions. The Lo et al. functional score represents relative enzyme activity (high score = retained function, low score = loss of function), while AlphaMissense scores increase with predicted pathogenicity. As expected, a negative correlation was seen (Pearson $r = -0.446$, Spearman $\rho = -0.472$, $n = 376$, Figure 4), which indicated that variants with high AlphaMissense scores corresponded to those with reduced functional output in the yeast assay. When coloring by ClinVar classification, pathogenic and likely pathogenic variants cluster in the upper-left quadrant (high AlphaMissense, low functional score), while benign and likely benign variants tend toward lower AlphaMissense scores and higher functional retention. VUS variants are distributed across the entire range, again reflecting uncertainty.

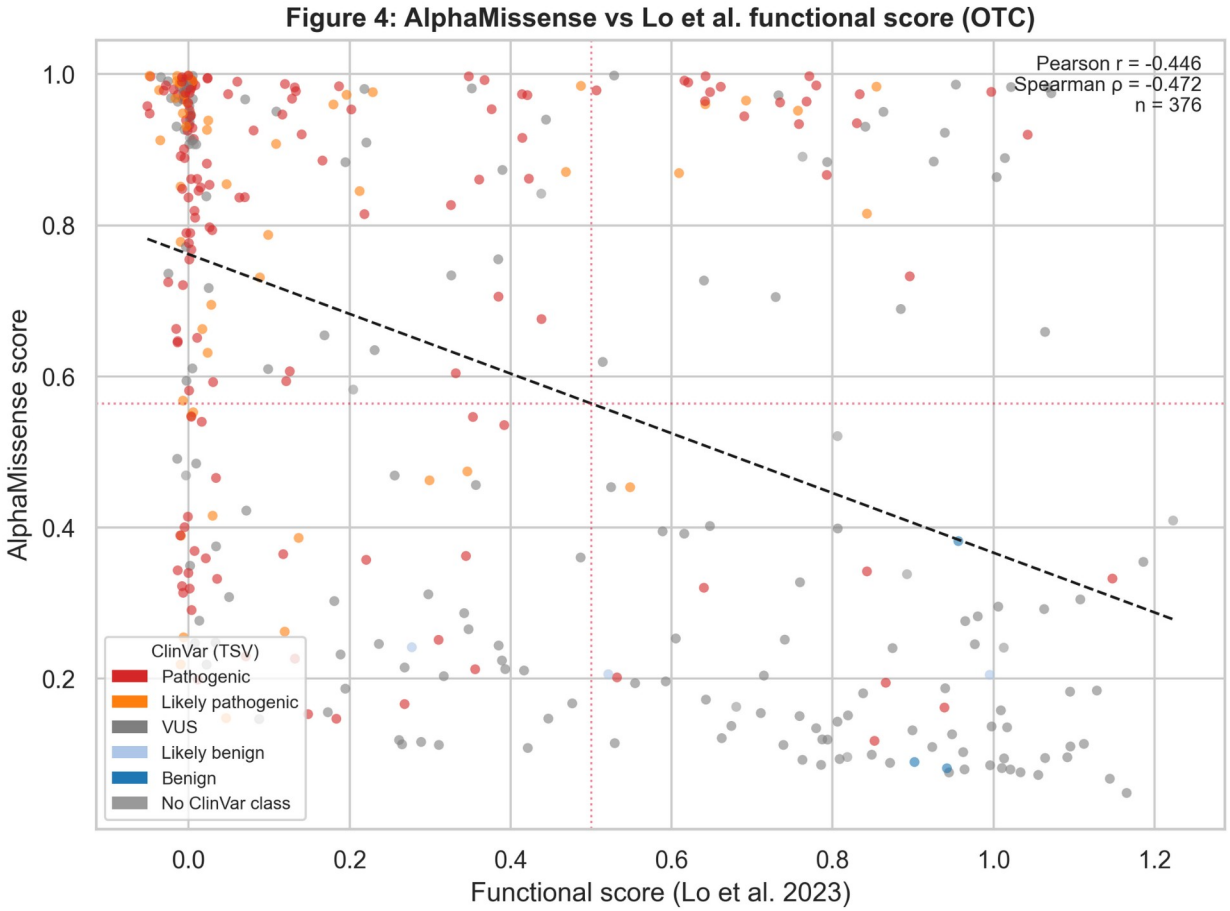


Figure 4. *AlphaMissense score versus Lo et al. (2023) functional score for OTC variants. Each point represents one OTC variant. X-axis: relative yeast growth score from Lo et al. (2023), high scores indicate retained enzyme function. Y-axis: AlphaMissense pathogenicity score. Points are colored by ClinVar classification (gray = no ClinVar record). Dashed black line: linear regression. Pink dotted lines: reference thresholds (AM=0.564, functional score=0.5).*

Interestingly, a large proportion of variants with functional scores near zero have AlphaMissense scores below 0.564, and several variants at functional scores approaching 1.0 still carry elevated AlphaMissense scores as well. This variation shows the limits of a sequence and structure-based predictor applied to the yeast functional system, especially with the complexity of partial loss-of-function variants shown in Lo et al.

Free energy change predictions (ΔΔG in kcal/mol) computed by DynaMut2 were compared against Lo et al. functional scores for 373 overlapping OTC variants (Figure 5). A significant but weaker positive correlation was observed (Pearson r = 0.286, Spearman ρ = 0.310, n = 373). Here, a positive correlation is expected because more negative ΔΔG values (destabilizing) correspond to lower functional scores.

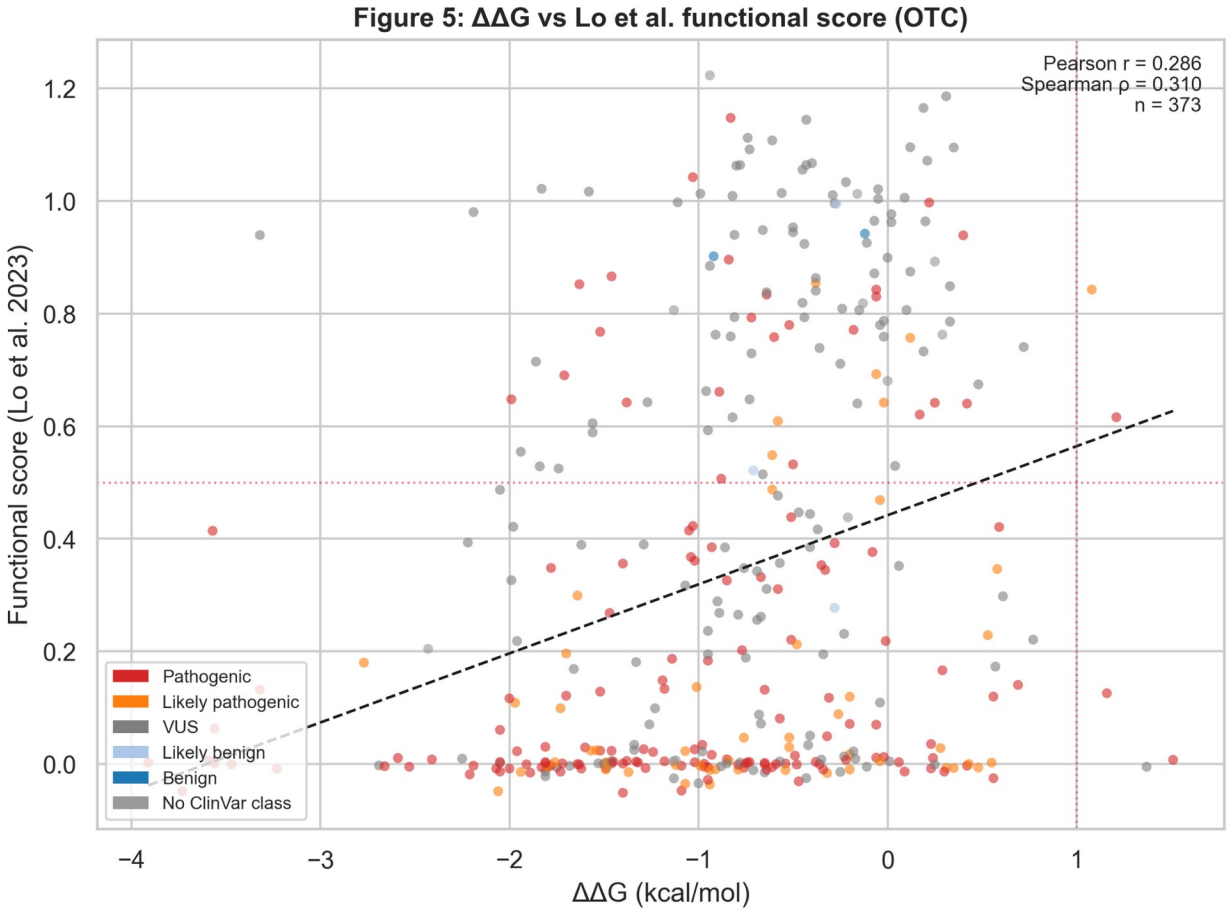


Figure 5. *DynaMut2 ΔΔG versus Lo et al. (2023) functional score for OTC variants. Each point represents one OTC variant with both a DynaMut2 ΔΔG prediction and a Lo et al. functional score. X-axis: ΔΔG (kcal/mol), positive values indicate thermodynamic destabilization. Y-axis: relative yeast growth score. Points colored by ClinVar classification (gray = no ClinVar record). Dashed black line: linear regression. Pink dotted vertical line: ΔΔG = 1.0 kcal/mol destabilizing threshold.*

The weaker correlation of ΔΔG relative to AlphaMissense (r = 0.286 vs 0.446) suggests that thermodynamic destabilization of the folded protein might not be the dominant mechanism by which the majority of OTC pathogenic variants impact the gene. The majority of DynaMut2-predicted ΔΔG values cluster between −2 and 0 kcal/mol, with a few variants over +1 kcal/mol. This proposes that most OTC substitutions are predicted to be only modestly or neutrally destabilizing. The performance of all three computational tools against experimental data is summarized in Table 2.

Table 2: OTC predictive performance

Predictor	Comparison Dataset	Pearson r	Spearman ρ	n	Pearson p-value	Spearman p-value
AlphaMissense	Lo et al. 2023 OTC functional score	-0.446	-0.472	376	8.10e-20	2.67e-22
DynaMut2 ΔΔG	Lo et al. 2023 OTC functional score	0.286	0.310	373	1.85e-08	9.03e-10
REVEL	Lo et al. 2023 OTC functional score	-0.239	-0.214	376	2.87e-06	2.85e-05
AlphaMissense	DynaMut2 ΔΔG (OTC inter-predictor agreement)	-0.213	-0.219	454	4.64e-06	2.36e-06

Table 2. *Predictive performance of computational tools against experimental OTC functional data. Pearson r and Spearman ρ correlations are reported against Lo et al. (2023) relative yeast growth scores. AlphaMissense scores are negatively correlated by design (higher score = greater predicted functional impact = lower yeast growth). ΔΔG values are from DynaMut2 (positive = destabilizing).*

To characterize the relationship between sequence-based functional prediction and thermodynamic stability, OTC variants with both AlphaMissense and DynaMut2 scores were

classified into four categories using thresholds of AM = 0.564 and ΔΔG = 1.0 kcal/mol (Figure 6).

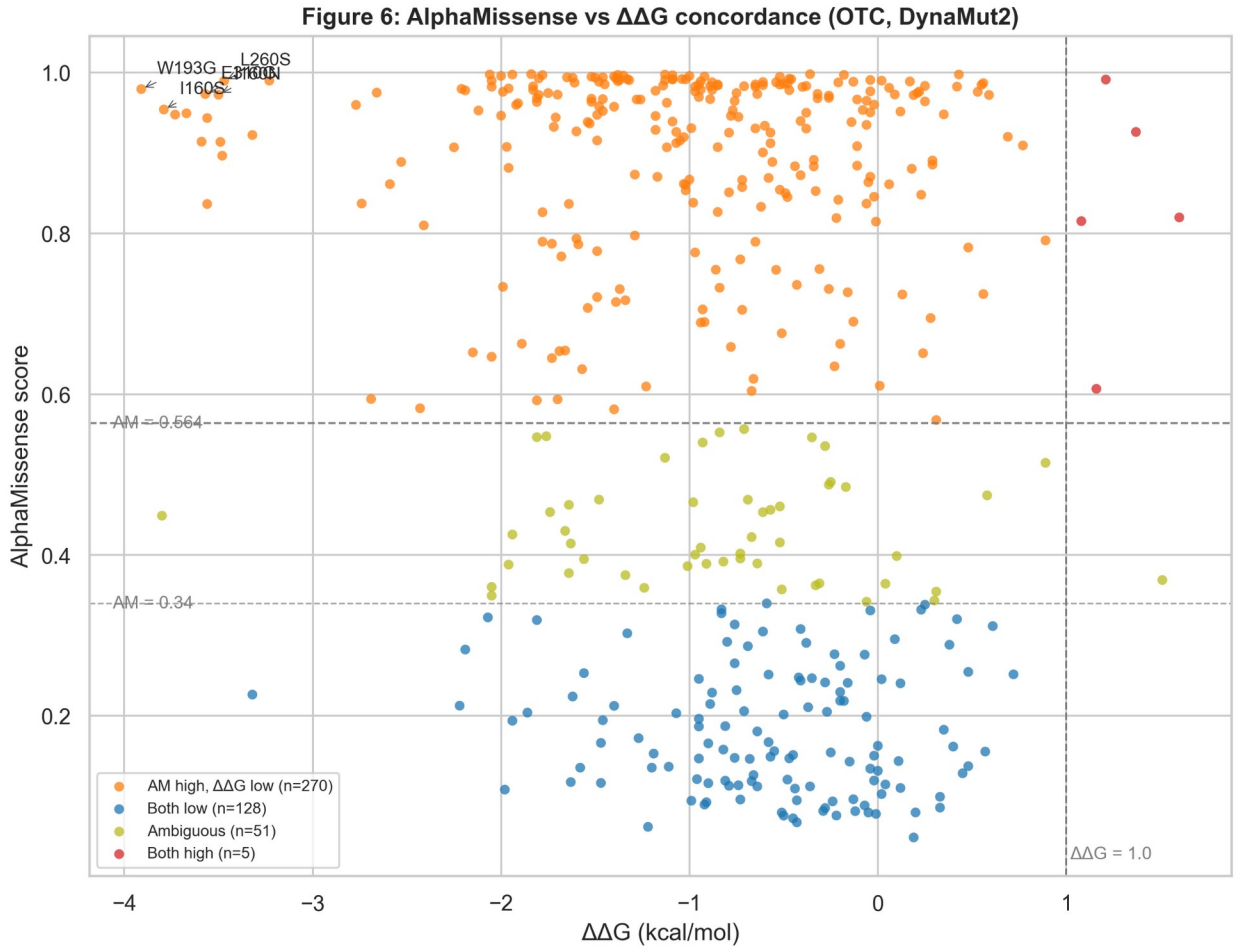


Figure 6. *AlphaMissense score vs. DynaMut2 ΔΔG concordance analysis for OTC variants. Each point represents one OTC variant with both scores. Points are colored by concordance category: orange = AM high / ΔΔG low (n=270); blue = both low (n=128); olive = ambiguous (n=51); red = both high (n=5). Gray dashed lines indicate classification thresholds (AM=0.564, AM=0.34, ΔΔG=1.0 kcal/mol).*

The largest category was AM high / ΔΔG low (n=270, 59.5%), being made up of variants for which AlphaMissense predicts high functional impact, but DynaMut2 predicts little or no global thermodynamic destabilization. The second largest category was both low (n=128, 28.2%), which was made up of variants predicted as neutral by both tools. Only 5 variants (1.1%) were

classified as both high (high AlphaMissense, destabilizing by ΔΔG). This reflects that perhaps classical unfolding is a relatively uncommon mechanism in OTC pathogenicity. The ambiguous category (n=51, 11.2%) contained variants for which the two tools gave intermediate or conflicting signals.

The large proportion of AM high / ΔΔG low variants supports the interpretation that most disease-causing OTC missense mutations act via disorder of active site geometry or substrate binding rather than through protein unfolding as a whole, which ΔΔG would have accounted for. This is consistent with the structure of OTC, in which substrate detection involves a constellation of catalytic residues, and mutations affecting local geometry or electrostatic relations can hinder activity without completely destabilizing the trimeric fold of OTC. Several disagreeing variants (W193G, L260S, E313N, I130N, and I160S) were labeled in Figure 6 and represent candidates for extra experimental investigation, as their computational profiles suggest mechanisms not captured by either predictor.

A direct comparison of all three data sources was performed for the 18 OTC variants experimentally characterized by Micheloni et al. (2025) through kinetic assays (Figure 7). AlphaMissense scores, ΔΔG values, Lo et al. functional scores, and Micheloni experimental classifications were aggregated into a single heatmap.

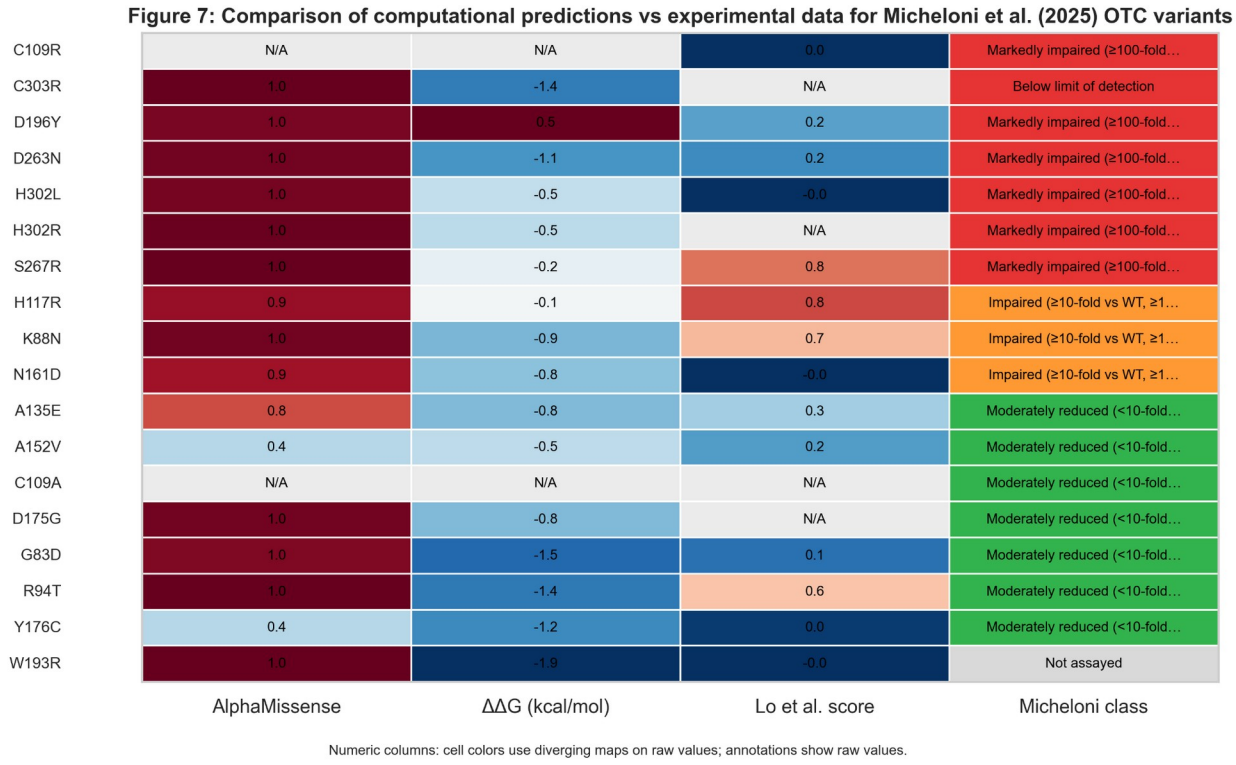


Figure 7. *Heatmap comparison of computational predictions and experimental classifications for 18 OTC variants from Micheloni et al. (2025). Rows represent individual variants sorted by Micheloni's experimental classification (markedly impaired → impaired → moderately reduced). Columns show AlphaMissense score, DynaMut2 ΔΔG (kcal/mol), Lo et al. (2023) functional score, and Micheloni et al. (2025) classification. Numeric columns use diverging color maps centered at 0.5 (AlphaMissense, Lo et al.) or 0 (ΔΔG). Micheloni classification column uses categorical colors (red = markedly impaired, orange = impaired, green = moderately reduced, gray = not assayed). N/A indicates data not available for that variant-predictor pair.*

For variants classified as markedly impaired (≥100-fold reduction in catalytic efficiency) by Micheloni et al. (D196Y, D263N, H302L, H302R, S267R, and C303R), AlphaMissense scores were uniformly near 1.0, showing a strong correlation between the kinetic assay results and sequence-based prediction. K88N and N161D, classified as impaired (≥10-fold reduction), carried AlphaMissense scores of 1.0 and 0.9. Moderately reduced variants showed more

variability: A152V and Y176C had AlphaMissense scores of only 0.4, suggesting that AlphaMissense underestimates the impact of these substitutions relative to the enzymatic data.

ΔΔG values for these variants were mostly negative, consistent with the pattern seen in prior ΔΔG analysis, and did not cleanly separate the Micheloni functional classes. Additionally, C109R and C109A didn't have AlphaMissense scores due to their absence from the AlphaFold DB at these specific substitutions, representing a gap in the AlphaMissense resource. The Lo et al. functional score was unavailable (N/A) for several variants, showing the limitation that even the most comprehensive functional assay to date does not cover every known clinical variant.

To assess whether the position of a variant within the OTC structure influences its predicted pathogenicity, variants were sorted by Cα–Cα distance to the nearest annotated active site residue (R92, R141, Q171, D263, S267, R330) using the 1C9Y crystal structure (Figure 8). Variants were further separated by AlphaMissense score range (AM >0.8, 0.4–0.8, <0.4).

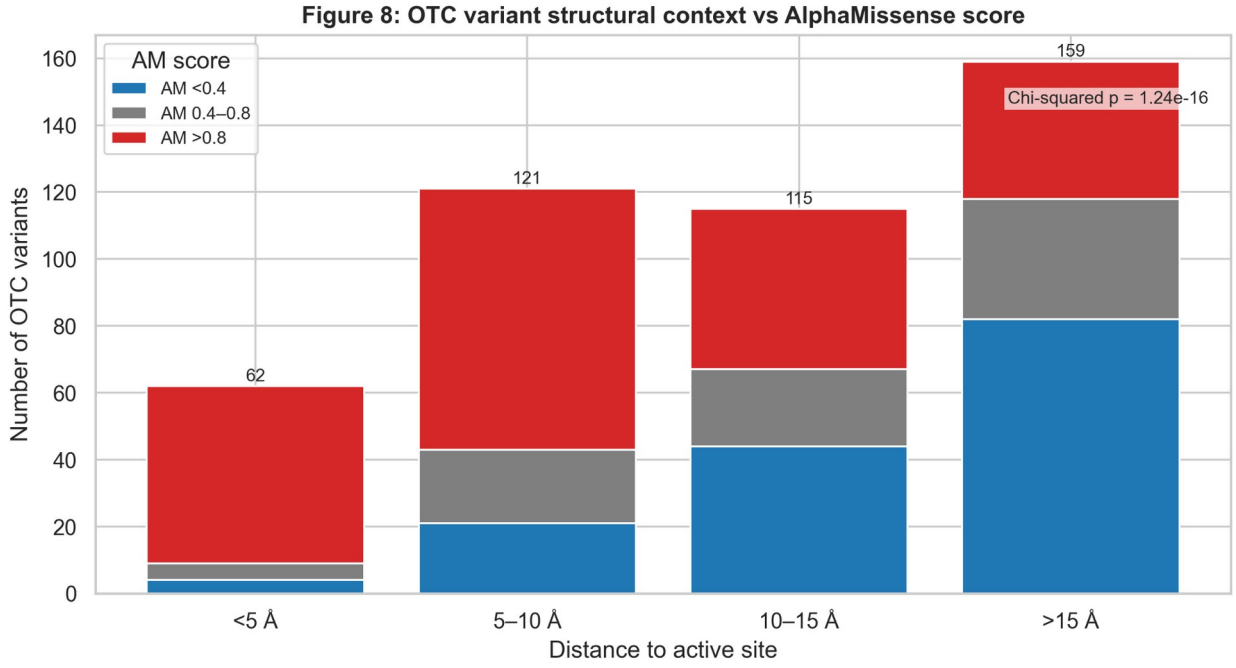


Figure 8. *OTC variant AlphaMissense score distribution by distance to active site. Stacked bar chart showing the proportion of OTC variants in each distance bin (<5 Å, 5–10 Å, 10–15 Å, >15 Å from the nearest active site*

residue), colored by AlphaMissense score range (red = AM >0.8; gray = AM 0.4–0.8; blue = AM <0.4). Active site residues used: R92, R141, Q171, D263, S267, R330 (1C9Y crystal structure).

A significant association between distance to the active site and AlphaMissense score distribution was observed (χ^2 $p = 1.24\times10^{-16}$). Among the 62 variants within 5 Å of the active site, the majority carried AM scores above 0.8, with few falling below 0.4. As distance increased, the proportion of low-AM variants grew as well. In the >15 Å bin (n=159), low-AM variants (blue) comprised the largest single component, while high-AM variants (red) shrank proportionally. The 5–10 Å bin (n=121), which had the highest total count of variants, had high-AM variants still dominant. These findings demonstrate that AlphaMissense scores are not uniformly distributed across the protein but rather are heavily influenced by the active site, supporting the model that active site proximity is another driver of predicted variant pathogenicity in OTC.

Discussion

This study aims to showcase a multi-dimensional computational framework with sequence-based pathogenicity prediction, thermodynamic stability analysis, structural context, and community-supported clinical classifications that can characterize missense variants across all eight urea cycle genes at a large and accessible scale. The OTC validation analysis showed that active site disruption, rather than global protein destabilization, is the predominant mechanism by which OTC variants become pathogenic. This is supported by the weak ΔΔG correlation with functional outcomes ($r = 0.286$) relative to AlphaMissense ($r = -0.446$), the concordance analysis showing 59.5% of scored OTC variants carry high AlphaMissense scores despite ΔΔG values below the destabilizing threshold, and the significant relation between active site proximity and predicted pathogenicity (χ^2 $p = 1.24 \times 10^{-16}$).

This interpretation is consistent with the known structure of OTC. The enzyme functions as a homotrimer, with each active site formed at the boundary of the carbamyl phosphate binding domain and the ornithine binding domain, and then connected by a flexible hinge. The catalytic mechanism depends on an alignment of residues that orient substrates and regulate proton transfer: R92, R141, D263, R330, and the SMG loop (S267). A missense substitution at or near any of these residues can detrimentally affect catalytic activity through the repositioning of a side chain, disruption of a hydrogen bond, or change in electrostatic potential at the active site, all without destabilizing the entire trimeric fold. This can be contrasted with variants in simpler enzymes, where destabilization and loss of function are more connected.

The significant association between AlphaMissense scores and ClinVar classifications (Kruskal-Wallis $p = 1.58 \times 10^{-23}$) further demonstrates that the precomputed scores continue to

align with the collective judgment of diagnostic laboratories submitting to ClinVar. This supports the utility of AlphaMissense as a first-pass prioritization tool for the UCD variant interpretation workflow. However, with the VUS distribution spanning nearly the full 0–1 score range, VUS resolution cannot rely on AlphaMissense alone. The multi-dimensional approach, combining AlphaMissense, ΔΔG, REVEL, structural proximity, ClinVar submission count, and LitVar paper linkage, is more accurate in identifying VUS than any single predictor in isolation.

The negative correlation between AlphaMissense and Lo et al. functional scores (Pearson $r = -0.446$, Spearman $\rho = -0.472$) matches benchmarks reported in the original AlphaMissense publication. Cheng et al. (2023) reported an area under the receiver operating characteristic curve (AUC) of 0.940 for distinguishing pathogenic from benign variants on ClinVar data. While AUC is not directly comparable to Pearson r for a continuous functional score, the moderate correlation observed here makes sense given that Lo et al.'s yeast growth assay records a continuous spectrum of function rather than a binary classification. Weaker predictors such as PolyPhen-2 and SIFT showcase correlations in the $r = 0.3–0.4$ range on similar continuous functional datasets (Fowler and Fields, 2014), putting AlphaMissense's performance on OTC at the higher end of published benchmarks for sequence-based tools.

The weaker performance of DynaMut2 ΔΔG ($r = 0.286$) is also consistent with published data. Rodrigues et al. (2021) report that DynaMut2 achieves a Pearson correlation of about 0.72 against experimental ΔΔG measurements from thermodynamic databases. The task here, however, is correlating ΔΔG against a functional growth score rather than predicting thermodynamic ΔΔG against calorimetric ground truth. The weaker correlation here is similar to what was shown by the concordance analysis: for OTC, pathogenicity is largely driven by active site disruption rather than destabilization of the entire molecule, so a thermodynamic predictor such as ΔΔG is working outside its primary domain of applicability for the majority of variants.

The gene-level variation in AlphaMissense score distributions is another important and novel observation for the urea cycle gene family. The low median score for NAGS is especially notable. Unlike the other five enzymes, NAGS produces N-acetylglutamate, an allosteric activator of CPS1, rather than actually catalyzing a step in the urea cycle itself. As a regulatory enzyme rather than a direct pathway enzyme, NAGS may tolerate a wider range of missense mutations without being likely to cause disease. This is consistent with its lower overall pathogenicity scores and the relatively lower number of ClinVar submissions compared to OTC and CPS1. Supporting this is the clinical observation that NAGSD is one of the rarest and most pharmacologically treatable UCDs.

The validation against Micheloni et al. (2025) revealed that computational tools perform best at both ends of a functional spectrum (clearly damaging vs. clearly neutral) and struggle most with partial loss-of-function variants and VUS. AlphaMissense assigned scores of 1.0 to all variants Micheloni classified as markedly impaired, but assigned scores of only 0.4 to A152V and Y176C, both of which had moderate enzymatic reduction demonstrated in the kinetic assays. This is also seen in observations by Fowler and Fields (2014) that sequence-based methods tend to underestimate the impact of substitutions at partially tolerant positions.

There are some limitations that should be kept in mind when interpreting these findings. First, the variants submitted to ClinVar are those observed in clinically affected patients or identified through diagnostic testing, misrepresenting the actual amount of mutations present. This means the AlphaMissense score separation between pathogenic and benign ClinVar groups (Figure 3) likely reflects a somewhat favorable comparison environment. Second, the Lo et al. yeast functional assay, which might be the most complete OTC dataset available, is not a direct measure of human enzyme activity. In that study, growth-based assays in a complementation

system measured the ability of a variant to restore function in a yeast model, which might not perfectly match the kinetics, folding environment, or cellular regulation of human OTC. The functional score values at or below zero in Figures 4 and 5 reflect variants indistinguishable from loss-of-function in that system, but some of these variants might only have detectable residual activity via direct kinetic measurement. Third, the ASL structural analysis is based on the 1AOS crystal structure at 4.2 Å resolution, which is not ideal for reliable side-chain-level analysis. Active site distance calculations for ASL variants were thus flagged as lower-confidence in the platform, and structural annotations for this gene should be looked at with care while waiting for a higher-resolution crystal structure to become available. Fourth, AlphaMissense and REVEL do not cover every variant. For example, C109R and C109A didn't have AlphaMissense scores, meaning gaps exist in the resource, which might affect a small but non-trivial fraction of clinically reported variants. Future updates to the AlphaMissense dataset should improve this gap. Finally, the literature extraction pipeline recovers variants from PubMed abstracts, not variants only mentioned in full text and supplementary data. The low literature-extracted counts relative to ClinVar (for example, 0 literature variants for ASL) do not reflect the true concentration of variant mentions in literature, but rather the dependence on abstracts instead of full-length text. Integration of full-text mining through PubTator or LitVar would significantly help this and increase variants pulled from texts, but not solve the issue entirely.

Future directions for this platform are planned. First, the multi-dimensional scoring framework validated here for OTC will be extended to the remaining seven urea cycle genes as DynaMut2 computations and structural annotations are completed. This will allow for cross-gene comparisons of mechanisms and computational performance. Rummagene analysis, by querying the gene set across millions of PMC supplementary tables, may also introduce unexpected

biological co-expression contexts for the urea cycle genes, such as the known association of ASS1 with arginine biosynthesis in cancer cells, and will also be incorporated into the platform's literature discovery features.

The integration of LitVar 2.0 will allow for variant detail pages with papers that abstract extraction misses, especially for variants described only in case reports or clinical genetics databases rather than journals. As the community of users grows and the like/dislike/comment interaction data accumulates, the popularity-based sorting system will provide an increasingly more useful signal for identifying the most impactful publications in each gene's literature.

Longer-term, the platform can be positioned to integrate retrieval-augmented generation (RAG) agents of the type demonstrated by Craig and Drăghici (2024), enabling natural language queries against the curated literature base. Such an agent would be constrained to the UCD literature given and aware of variant-level computational scores, serving as an on-demand clinical decision support tool for geneticists encountering an unfamiliar OTC or CPS1 variant in a newly diagnosed patient.

Conclusion

This project was done to present the Urea Cycle Research Hub, an accessible web platform for researchers integrated with automated PubMed literature retrieval, ClinVar variant import, and multi-dimensional computational scoring for all eight urea cycle genes. A total of 2,341 unique missense variants were catalogued, with AlphaMissense, REVEL, and DynaMut2 scores for each, and structural diagrams were embedded in the browser interface via experimental crystal structures.

OTC analysis showed that AlphaMissense scores significantly correlated with the experimental functional results from a 1,570-variant high-throughput assay (Pearson $r = -0.446$, $p < 0.001$) and are also strongly correlated with established ClinVar clinical classifications (Kruskal-Wallis $p = 1.58 \times 10^{-23}$). DynaMut2 thermodynamic analysis showed a weaker but still significant association with the same functional data ($r = 0.286$), and concordance analysis demonstrated that the likely mechanism of OTC variant pathogenicity is active site disruption rather than entire protein destabilization. This was supported by the significant relationship between active site proximity and predicted pathogenicity (χ^2 $p = 1.24 \times 10^{-16}$). These findings were further supported by comparison with kinetically characterized variants from Micheloni et al. (2025).

Together, these results congregate to show that the integration of a sequence-based, thermodynamic, and computational framework within an accessible, community-based knowledge platform will provide meaningful infrastructure for researchers to prioritize urea cycle variants for experimental validation. While computational predictions currently can’t replace direct functional characterization, the combination of multiple, independent predictors can be the knowledge base for the many thousands of variants that might never be

experimentally characterized at scale. Overall, this platform serves as a foundation for the UCD research community to collectively build that understanding.

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